



job portal sysytem

Bachelors of computer application (Tribhuvan Vishwavidalaya)



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Tribhuvan University
Faculty of Humanities and Social Sciences

ONLINE JOB PORTAL

A PROJECT REPORT

Submitted to
Department of Computer Application
Birendra Multiple Campus

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by

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Supervisor's Recommendation

I hereby recommend that this project prepared under my supervision by **Aadesh Bhandari** and **Jagadis Pokharel** entitled “**ONLINE JOB PORTAL**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

SIGNATURE

Bindu Aryal

SUPERVISOR



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LETTER OF APPROVAL

This is to certify that this project prepared by **Jagadis Pokhrel** and **Aadesh Bhandari** entitled “**ONLINE JOB PORTAL**” in partial fulfillment of the requirements for the degree of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

SIGNATURE of Supervisor	SIGNATURE of HOD/ Coordinator
SIGNATURE of Internal Examiner	SIGNATURE of External Examiner
Internal Examiner	External Examiner

Abstract

In this competitive era, the education among the people is so increasing that the jobs for them are now decreasing. The companies even want the people who are best in their fields. At that time, it becomes difficult to find the people who are intelligent enough to be hired. Thinking about these problems, one can think about the process which can handle this process and make the work less complex. This project is about the recruitment process which is done online. The recruitment process here is handled by the system. This project will allow the person to apply for a job in the company for the interested vacancy. The person will be having the account after registration and will be able to see the latest job vacancies. The vacancies will be posted by the admin. Being an authorized admin he can publish vacancy details and can search no of Employees on portal and can select candidates on the basis of the key skill which provided during registration.

Acknowledgement

This project would not have been possible without the help and guidance of many individuals despite the huge efforts from our side. We would like to extend our sincere thanks to all. We are indebted to **BIRENDRA MULTIPLE CAMPUS** and its members for the support and supervision and also for providing the necessary information regarding this project.

Also, we would like to thank our project guide Mrs. **Bindu Aryal** and Head of department Mr. **Sobaraj Poudel** for their kind co-operation and guidance which has helped us in the completion of the project. We give our special thanks and kind appreciation to our seniors, colleagues and a few individuals for their help during our project.

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List of Abbreviations

HTML: Hypertext markup language

CSS: Cascading Style Sheet

PHP: Hypertext Preprocessor

DFD: Data Flow Diagram

ER: Entity Relationship

SQL: Structured query language

CV: Curriculum vitae

LCD: Liquid crystal display

RAM: Random access memory

HDD: Hard disk drive

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Chapter 1 Introduction

1.1 Introduction

A job portal, also known as a career portal, is a modern name for an online job board that helps applicants find jobs and aids employers in their quest to locate ideal candidates as per their experiences, qualification and so on. The system is to provide an alternate and convenient way for a person to find a job easily. Some government agencies, nonprofit organizations, universities and private businesses have their own job portals that applicants can access on the organization's website. This job portal can prove to be very helpful since it allows users of different profile to upload their CVs, search job on the basis of their qualification.

A job is very important factor of life, with job we can fulfill our desires and family needs. we also get self-respect, social status which make us the part of society. So this job portal helps individuals to be self-dependent and employed.

1.2 Problem Statement

In the context of Nepal, many job seekers have to face difficulties.

- A user doesn't get the proper information about the job.
- Online applications take too long to fill out and sometimes errors may occur.
- Candidates who apply online are often ignored.
- There is very long documentation process in job process so the process is time consuming.
- People don't get the proper job as per their qualification.

1.3 Objectives

This system can be used as an Online Job Portal for the Placements providing to the unemployed person who are seeking for a job placement. Job Seekers logging into the system and able to upload their information in the form of a CV. Admin logging in may also access/search any information put up by Job Seeker and contact them and acts as representative between recruiter company and job seeker.

1.4 Scope and Limitation

1.4.1 Scope

- Maintain the record of job seeker.
- Upload the customized job posting.
- Maintain the uploaded resumes.
- Maintain the job posting and various reports.
- Provides detail information about job.
- The process of registration is free.

1.4.2 Limitation

- This system is fully online based so internet is compulsory.
- Recruiter waste a lot of time with unqualified candidates.
- There is no fixed barrier so anybody can apply for any job.

1.5 Report Organization

This report document five chapters which describes the introduction of the built system. Chapter two defines and describes background study and overview of related existing systems and their pros and cons. Chapter three presents the system analysis and design including requirement analysis and feasibility analysis. In chapter five, conclusion, limitations, and future enhancement are briefly explained.

Chapter 2 Background Study and Literature Review

2.1 Background Study

The quality of people hired for a specific job is usually the key issue when measuring the effectiveness of the employment. However, in certain circumstances the speed of hiring is also an issue and also contribute to quality hiring The job recruitment or hiring employees process in the past years was based on advertisements in the newspapers and have a panel on the appointed day to interview applicants.

The traditional hiring of employees starts with the processing of application forms describe the job for each position verify application forms and lastly evaluate the best person for the job The whole process from advertisement to hiring a quality applicant takes a lot of time, effort and also has more weaknesses. The advertisement itself is costly especially when done through print media newspapers) therefore the publication of the job adverts will only last for a very short time and in that case few people will have seen the job vacancy. Moreover, submission of the application form requires physical presence of the applicant to hand it over and this could be inconvenience most applicants due to the fact that we belong to different geological locations in country. Thus, the traditional recruitment process is not being the best in today's competitive job market The rationale of a hiring process is to stretch out to potential employees and bring out the specific kind of required skills and experiences in the field organization. The question here is how this could be made efficient and possible. This question or problem can be solved by an online job portal system which is set to transform the way in which companies and other employers recruit their employees. Online job portal system is fundamental in removal of paper works and introduction of workflow systems that connect job seekers and employers and also stores data in reliable databases. [CITATION stu18 \l 1033]

2.2 Literature Review

Job Procurement: Old and New Ways

Job looking for more often than not includes distinctive approaches to search for jobs, for example, through close to home contacts, direct phone calls to businesses, job organization office, filtering on the web job postings, and so on[CITATION Mar10 \l 1033] . Prior to the Internet, turned out to be generally utilizes as a strategy for looking for jobs; jobseekers invested a heap of energy utilizing different strategies to search for job openings. Today, jobseekers utilize online techniques, which are extremely advantageous and spare a great deal of time. Galanaki[CITATION Ele02 \l 1033] records the accompanying strategies to be the conventional (old) routes for recruitment:

- Employment recruitment offices
- Job fairs
- Advertising in the broad communications, for example, papers
- Advertisement in TV and radio
- Management Consultants
- Existing worker contacts
- Schools universities or colleges understudies' administrations office

These old jobs looking for techniques are excessively moderate, distressing, testing and furthermore need quality[CITATION sai07 \l 1033] . What's more, the candidates need to consider the expense and the measure of time to get the data they need, and different arrangements they need to make. Securing all accessible position opening is a principle venture at in the job-chasing process. The Internet is presently an incredible asset that jobseekers can utilize. Today, numerous locales publicize job positions to be filled by individuals with specific skills in different fields. The Internet assumes an essential job in the region of human asset arranging and advancement. Most arranging and advancement organizations are currently utilizing PC innovation and the Internet for staff recruitment. It ought to be noticed that despite the fact that the Internet has

encouraged the procedure of job chasing, it has not supplanted the conventional techniques, totally

Chapter 3 System Analysis and Design

3.1 System Analysis

As we know that this project involves design and implementation of a software system regardless that is web-based, it will be important to mention and consider some modules used in software development. Some modules of software development are as follows:

- The waterfall approach: It represents activities in requirements, qualification, specifications, design, implementation, and testing. All these as separate processes.
- Incremental development: it involves a rapid development of the specification and then refined later for the customer.

After reading all these models the waterfall approach fits the development of this website. The main aim of using this approach is we can focus on each part of the model during development and come back to it if need be. The project can easily be broken down into different parts based on this model. This is the model we used to develop the job portal system. However, feedback loops will be allowed during the whole software development process. The model chosen for this project has to favor two developers for a project. Because we are the only one who we are going to implement this project. We find this model suitable for us. [CITATION Som19 \l 1033]

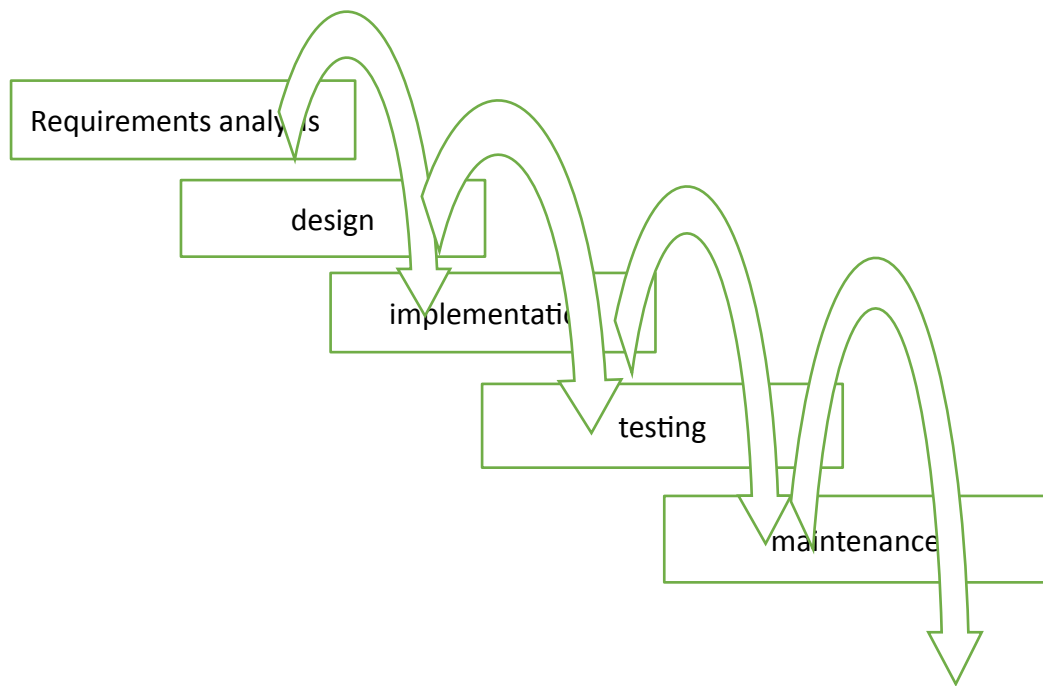


Figure 3-1 waterfall model

3.1.1 Requirement Analysis

Requirement analysis is done while developing a system and before implementing the system. It is necessary to analyze the whole system requirements. It is mainly categories into two parts. i.e. functional and non-functional requirements.

In any system there are functional and non-functional requirements to be considered while determining the requirement of the system. Functional requirement includes of the visible part of the system but the non- functional part describe how the system will do what is supposed to do.

I. **Functional requirements:** The functional requirement used in the project are mentioned below.

a) User

- i. User can register and login the system.
- ii. User can look at the job description.
- iii. User can apply to their suitable job
- iv. User can upload their cv.
- v. User can logout from the system

b) Admin

- i. Admin can add and delete job.
- ii. Admin can see the registered users.
- iii. Admin can view user who applied for jobs and their CVs.

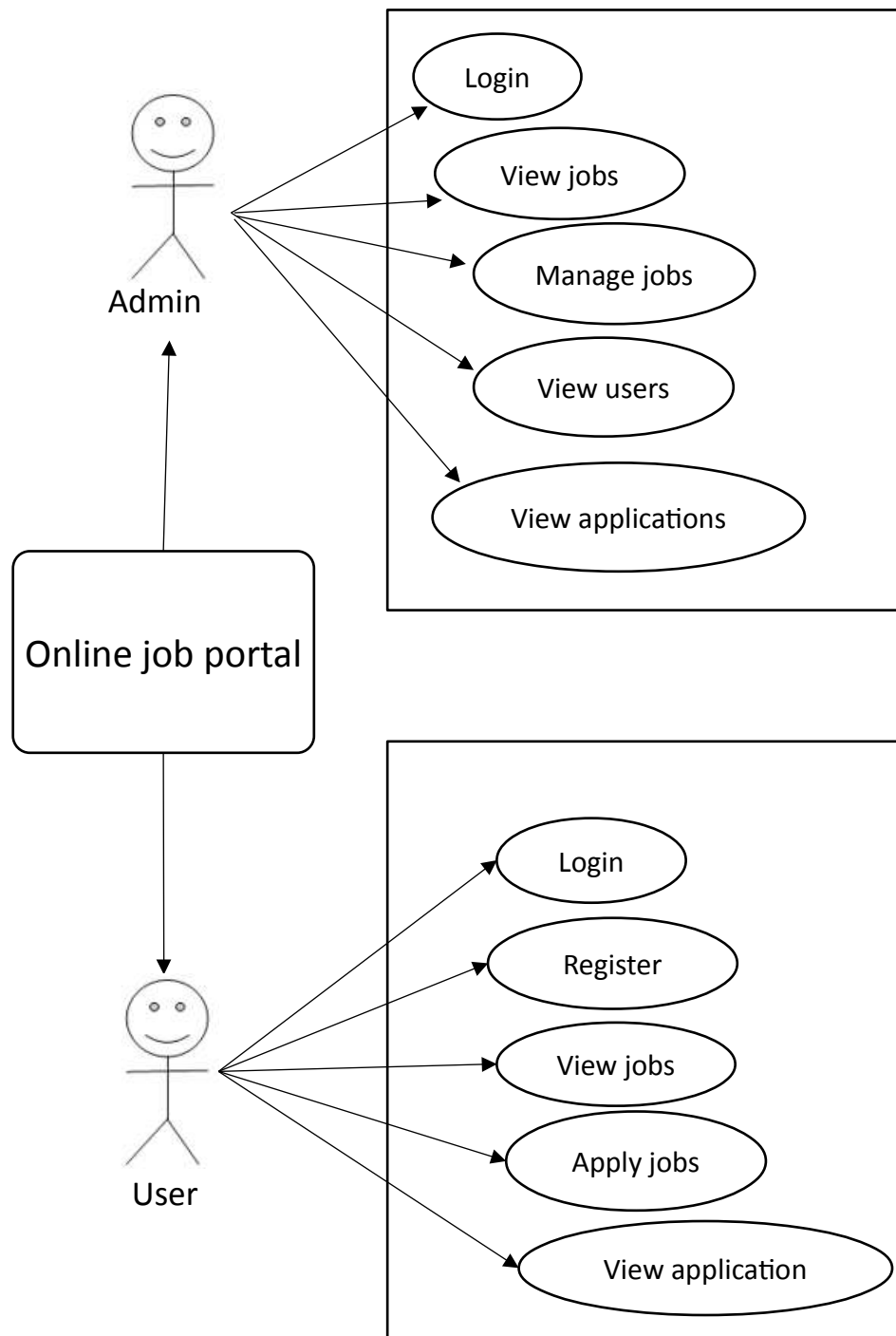


Figure 3-2 Used case diagram

II. **Non- functional requirements:** The non-functional requirement specifies how the system works. The non-functional requirements included in projects are:

Availability: It will be available online.

Security: This system will be secure and the user's information won't be available for others for users privacy.

Performance: This system will be designed for smooth performance with optimization and good response.

Reliability: It will be reliable for the user.

3.1.2 Feasibility Analysis

Technical feasibility: In order to design this system, it uses off-self and existing technologies, software and hardware so there is no technological hurdle to build this system

Table 3-1 Recommending the hardware parts

S.N.	Hardware	Specification
1.	monitor	LCD 13.5" Screen
2.	HDD	2 GB or above
3.	RAM	500mb or above
4.	Processor	Intel Celeron or above

The following software is used for the development of the system

- Vs code
- MySQL
- Apache [xampp]

And windows 10 is used as operating system as it is faster and reliable than other operating system.

i. Operational feasibility

These include the reliability, maintainability, usability, supportability. The proposed system is operationally feasible as it is reliable for all user. It is simple and easy due to simple user interface and its operational feasible.

ii. Economic feasibility:

Before the development of the system, the proposed system we should study whether the system is budget friendly or not. The project we are developing is within the cost estimation of the organization. The system development does not requirement of expensive hardware and software. Hence the project is said to be economically feasibility.

iii. Schedule feasibility:

The system that we developed is scheduling feasible as it does not require more time for the development. The data collection takes more time after the data is collected other development phase can be within a month. Gantt chart is a bar that provides of all information about how this system is schedule feasibility. The gantt chart is as shown below.

Week Task	2	4	6	8	10	12	14
Requirements analysis							
design							
implementation							
testing							
maintenance							
documentation							

Figure 3-3 Gantt chart

3.1.3 Data modeling (ER-diagram)

Data modeling is the process of creating a data model for the data to be stored in a database. This data model is a conceptual representation of Data objects, the associations between different data objects, and the rules. Data modeling helps in the visual representation of data and enforces business rules, regulatory compliances, and government policies on the data.[CITATION Dav21 \l 1033]

ER Diagram stands for Entity Relationship Diagram that displays the relationship of entity sets stored in a database. In other words, ER diagrams help to explain the logical structure of databases. ER diagrams are created based on three basic concepts: entities, attributes and relationships.

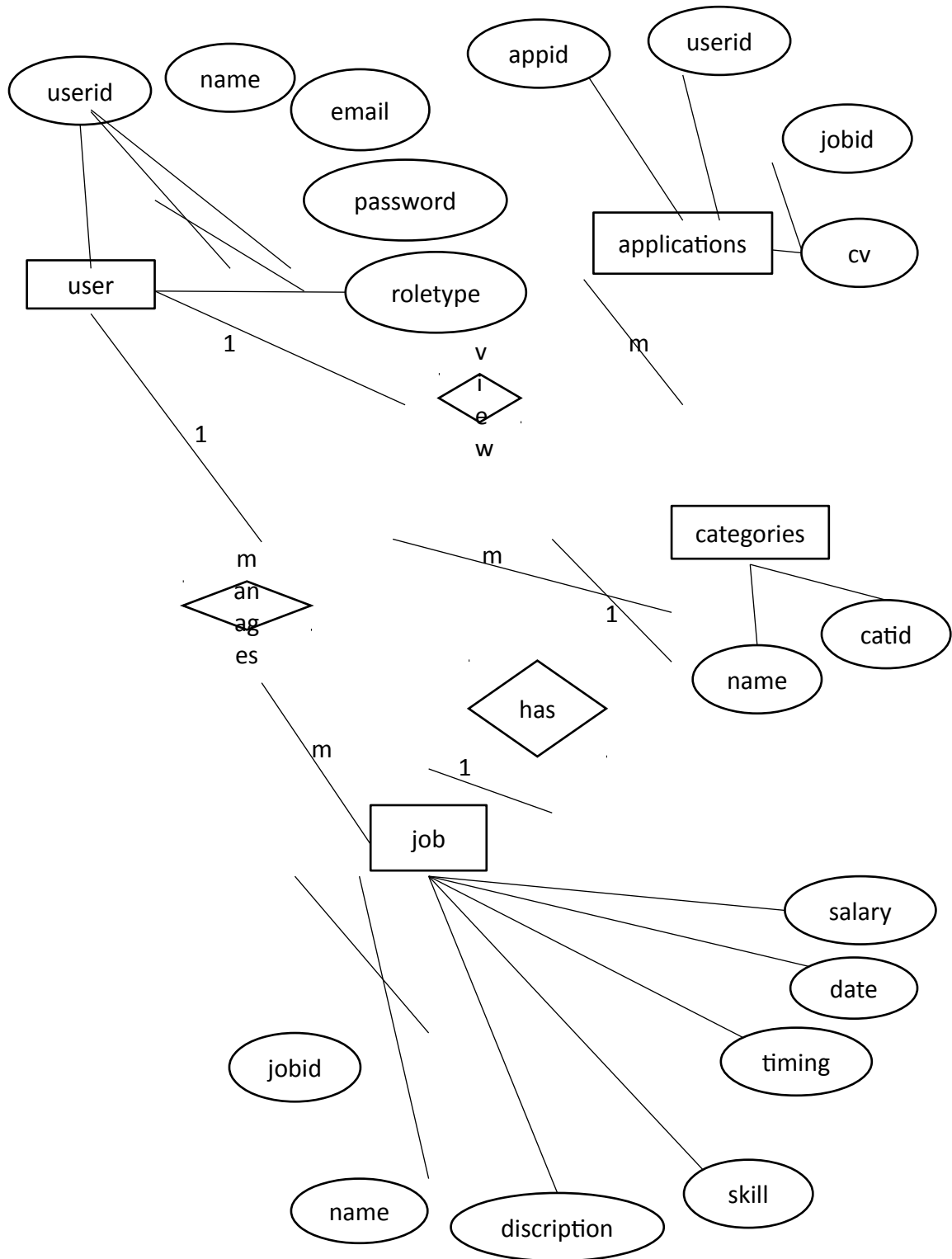


Figure 3-4 ER-diagram

3.1.4 Process Modeling (DFD)

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It can be manual, automated, or a combination of both.
[CITATION dat21 \l 1033]

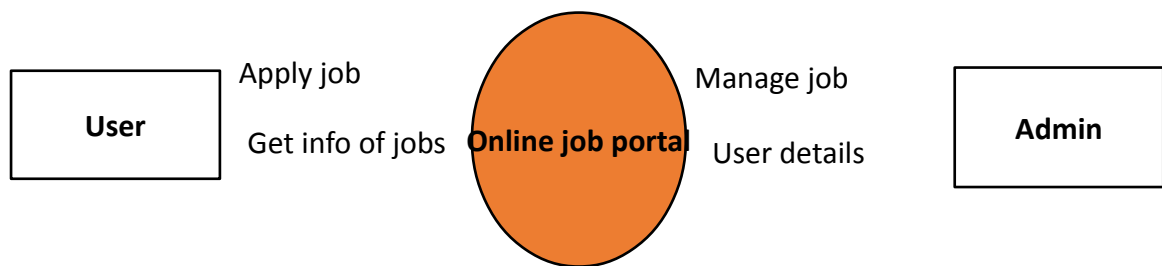


Figure 3-5 Level- 0 DFD

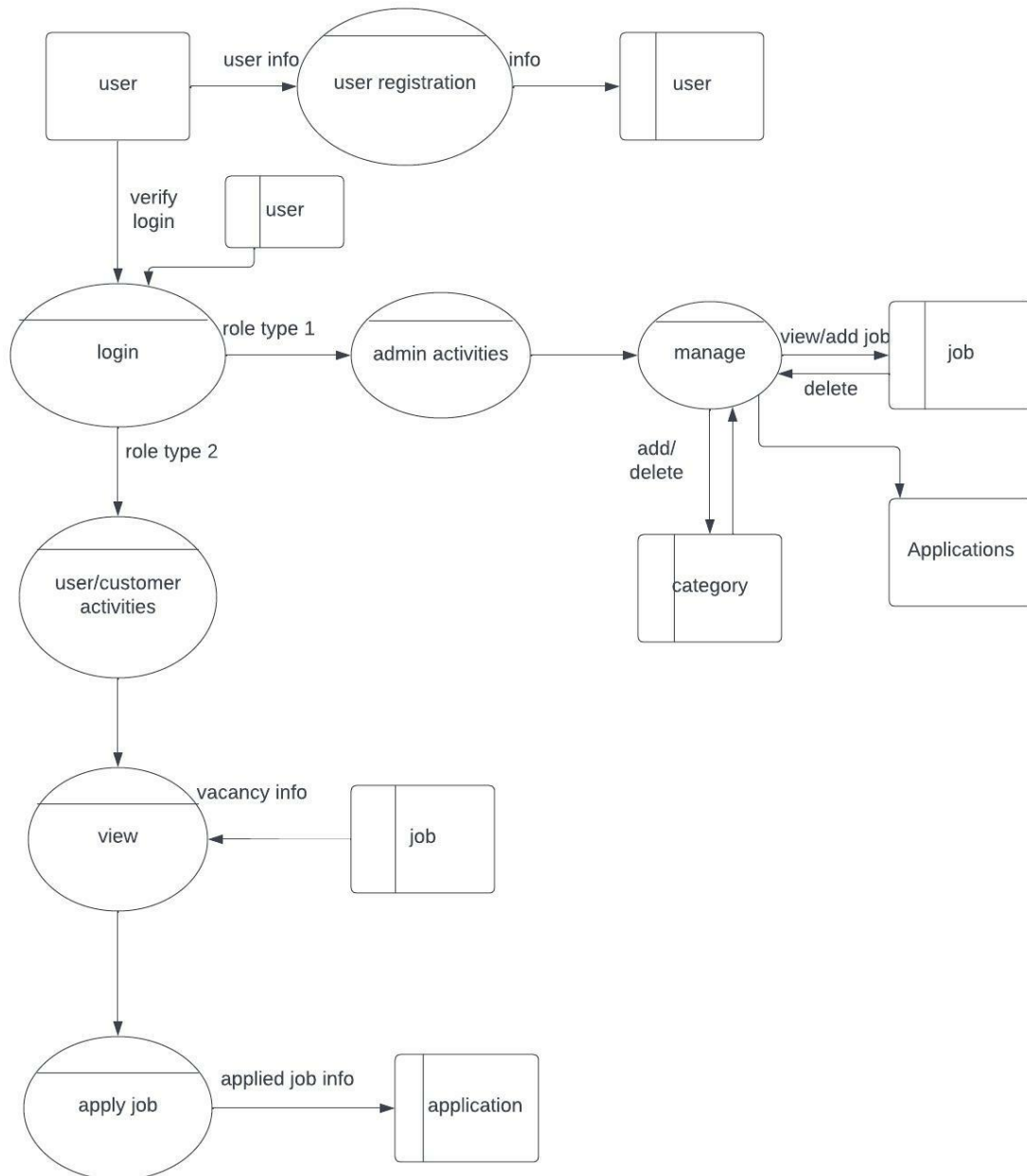


Figure 3-7Level -2 DFD

3.2 System Design

System design is the process of designing the elements of a system such as the architecture, modules and components, the different interfaces of those components and the data that goes through that system. In our system there are two modules. They are system module and admin module.[CITATION did18 \l 1033]

3.2.1 Architectural Design

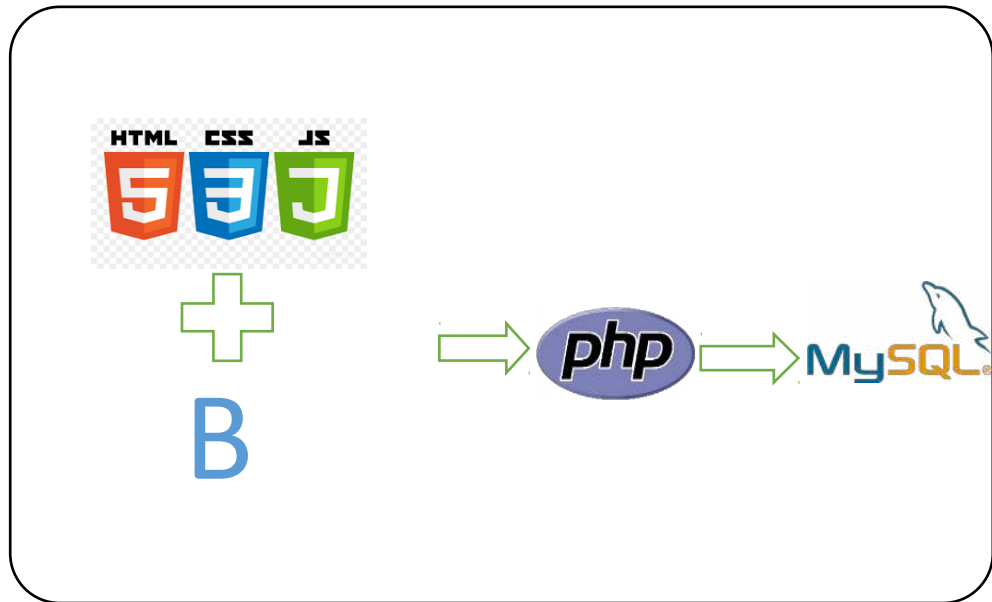


Figure 3-8 Architectural design

3.2.2 Database Schema Design

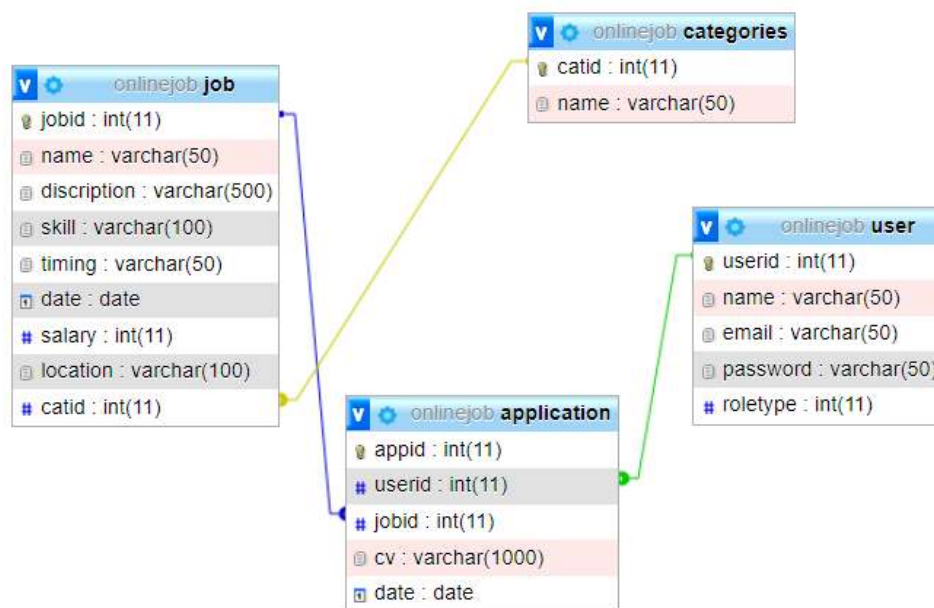


Figure 3-9 database schema design

3.2.3 Interface Design (UI Interface/Interface Structure Diagram)

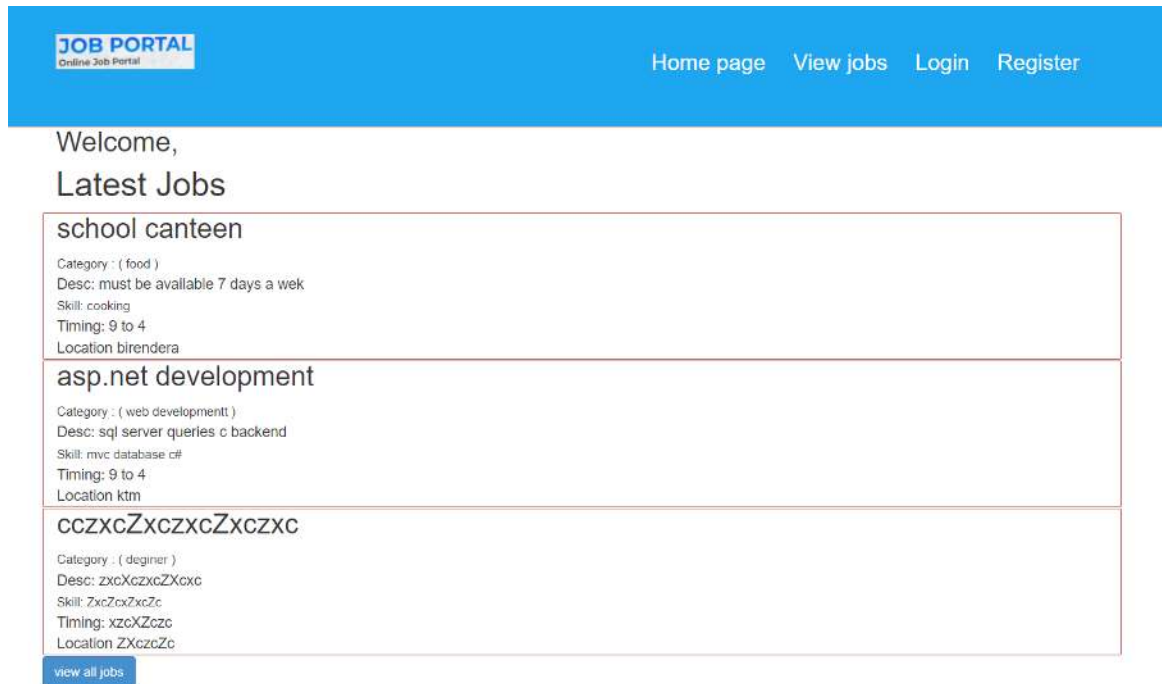


Figure 3-10 home page

Admin panel:

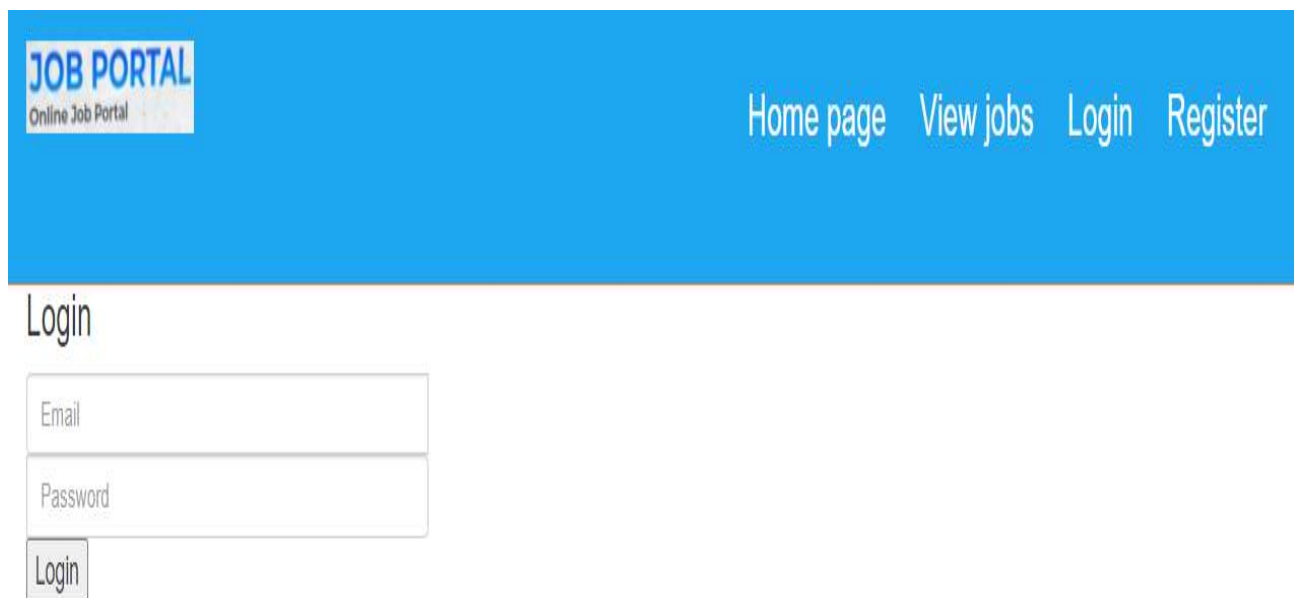


Figure 3-11 admin login panel

Add jobs

	Name
	Discription
	Skill
	Location
	Timing
	Salary
	teacher
Add Job	

ID	Name	Categories	Skill	Discription	salary	Time	Date	Action
14	school teacher	teacher	communication skill maths science	teacher required fot class 10	20000	9 to 4	2022-03-15	Delete
15	typing job	online	speed 40 wpm	its online	10000	10 - 5	2022-03-15	Delete
16	graphic designer	deginer	photoshop	designing job for flex	20000	9 - 5	2022-03-15	Delete
17	cczxcZxczxcZxczxc	deginer	ZxcZcxZxcZc	zxcXczxcZxcxc	24231243	xzcXZczc	2022-03-15	Delete
18	asp.net development	web developmentt	mvc database c#	sql server queries c backend	30000	9 to 4	2022-03-15	Delete

Figure 3-12 add job panel

Add categories



Name

Add Category

Update

ID	Name	Action
28	teacher	Edit Delete
29	online	Edit Delete
30	deginer	Edit Delete
31	web developmentt	Edit Delete
33	sfdsafdsad	Edit Delete
37	food	Edit Delete
39	dancer	Edit Delete
41	kjbbk	Edit Delete

Figure 3-13 add categories panel

All applications

ID	Job	Category	user	Date	CV
7	graphic designer	deglnr	aadesh	0000-00-00	view cv
12	typing job	online	aadesh	0000-00-00	view cv
8	school teacher	teacher	mausam	0000-00-00	view cv
9	asp.net development	web developmentt	mausam	0000-00-00	view cv
10	typing job	online	ram	0000-00-00	view cv
11	school teacher	teacher	keshav	0000-00-00	view cv
13	school canteen	food	jagadis	0000-00-00	view cv

Figure 3-14 view cv panel

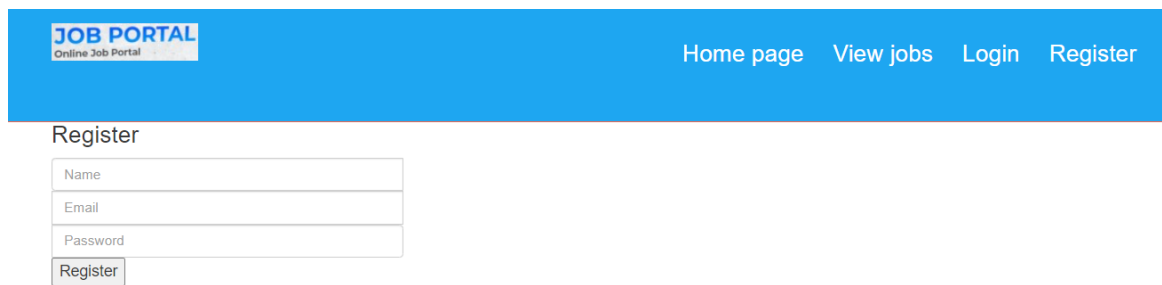
All users

ID	name	Email	pass
3	aadesh	aadesh@gmail.com	123
4	asd	asd@gmail.com	123
5	birendra	birendra@gmail.com	123
6	mausam	mausam@gmail.com	123
7	ram	ram@gmail.com	123
8	keshav	keshav@gmail.com	123
9	jagadis	jagadis@gmail.com	123

Figure 3-15 view user detail panel

XXX

User panel:



The screenshot shows the 'Register' section of the 'JOB PORTAL' website. The header is blue with the logo on the left and navigation links 'Home page', 'View jobs', 'Login', and 'Register' on the right. Below the header, the word 'Register' is displayed. The registration form consists of three input fields: 'Name', 'Email', and 'Password', stacked vertically. A 'Register' button is located at the bottom of the form.

JOB PORTAL
Online Job Portal

[Home page](#) [View jobs](#) [Login](#) [Register](#)

Register

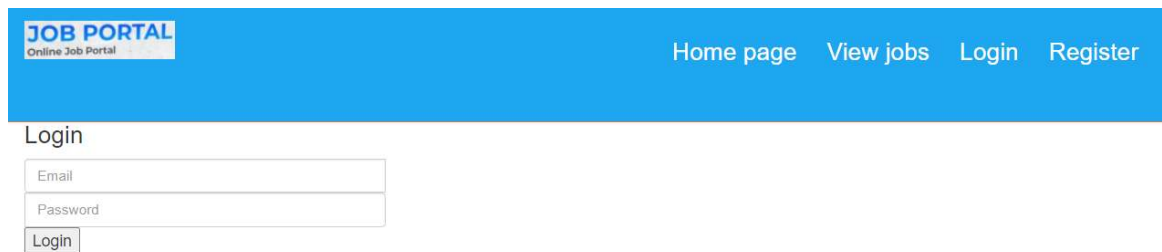
Name

Email

Password

Register

Figure 3-16 user register panel



The screenshot shows the 'Login' section of the 'JOB PORTAL' website. The header is blue with the logo on the left and navigation links 'Home page', 'View jobs', 'Login', and 'Register' on the right. Below the header, the word 'Login' is displayed. The login form consists of two input fields: 'Email' and 'Password', stacked vertically. A 'Login' button is located at the bottom of the form.

JOB PORTAL
Online Job Portal

[Home page](#) [View jobs](#) [Login](#) [Register](#)

Login

Email

Password

Login

Figure 3-17 user login panel

JOB PORTAL
Online Job Portal

[Home page](#)
[View jobs](#)
[Login](#)
[Register](#)

All Jobs

school canteen

Category : (food)

Desc: must be available 7 days a wek

Skill: cooking

Timing: 9 to 4

Location birendera

[More details](#)

asp.net development

Category : (web developmenttt)

Desc: sql server queries c backend

Skill: mvc database c#

Timing: 9 to 4

Location ktm

[More details](#)

CCZXCZxcZxcZxcZxc

Category : (deginer)

Desc: zxcXczxcZXcxc

Skill: ZxcZcxZxcZc

Timing: xzcXZczc

Location ZXczcZc

[More details](#)

graphic designer

Category : (deginer)

Desc: designing job for flex

Skill: photoshop

Timing: 9 - 5

Figure 3-18 user view job panel

Chapter 4 Implementation and Testing

4.1 Implementation

Implementation basically means the phases where the system is actually built. Firstly, all the information that we gathered is studied and analyzed and implemented system in operation for users. It is one of the most important phases of any project. Implementation usually consists of coding, testing, installation, documentation training and support. Different tools and technologies are used to develop this project which are already discussed above.

4.1.1 Tools Used (CASE tools, Programming languages, Database platforms)

The various tools that are used to develop this system are as follows:

Front-end: HTML, CSS, JavaScript, PHP

HTML is a syntax used to format a text document on the web. CSS is a style sheet language used for designing the looks and formatting of a document written in a markup language. Java script is a dynamic computer language. It is commonly used as part of web browsers, whose implementations allows client-side scripts to interacts with the user, control the browser, communicate asynchronously, and alter the document content that is displayed. This is used to create a popup windows displaying alerts in the system like “login successfully”, “purchase successfully” etc.

Back-end: PHP, MySQL, XAMPP

PHP is a server-side scripting language designed for a web development but also used as general-purpose programming language. PHP code is interpreted by a web server with a PHP processor module, which generates the resulting web pages: PHP commands can be embedded directly into an HTML source document rather than calling an external file to process data. PHP is free software released under PHP license. MySQL is world’s second most widely used relational database management system (RDBMS).

4.1.2 Implementation Details of Modules (Description of procedures/functions)

After design was made and the problem arising from the process were clarified and dealt with, It was time to start implementing the application. Implementing application of this scale requires lots of resources and explaining the whole implantation process will not be clarified in this paper. However major important aspects in implementation will be described. Some of the modules of job portal websites are listed below:

1. Header: It displays the logo of the website. It is used in navbar of the homepage.it is used in order to provide links to different pages of the website.
2. Register form: It is used to register new users to the website. It contains the text like email, username, password. The information entered is further stored to be used in login page.
3. Login form: it is used to provide a gateway to the website. It uses data like username, email, password from the register and give further access.
4. View job: By this section the user can see the job available in current days and select the suitable job to apply.
5. User module: It provides all the information related to the user.
6. Admin module: it provides information to the admin. It provides information like user detail, admin can add or remove job etc.

4.2 Testing

Testing is done to check the behavior of a complete and fully integrated software product based on the software requirement specification document. For an website and application or website to be deployed it has to be tested. Hence test cases will be written to test this website. There are amny types of testing procedures to check the working mechanism and correctness of the system.

4.2.1 Test Cases for System Testing

Admin login testing

Table 4-2 admin login testing

ID	Test case description	Test data	Expected result	Actual result	Pass/fail
TC1	Admin enters an valid email and pass	Email=admin@gmail.com Pass= 123	Admin login successfully	As expected	pass
TC2	Admin enters a invalid email and password	Email=admin@gmail.com Pass:1234	Invalid username or password	As expected	pass

Post job details by admin

Table 4-3 post job detail by admin

ID	Test case description	Test data	Expected result	Actual result	Pass/fail
TC3	Admin posts jobs with valid job datas	Name= teacher Description= full time Skill= math, science Location= bharatpur Timing= 10-4 Salary= 15000 Category=teacher	Record added	As expected	pass
TC2	Admin enters a invalid email and password	Name= teacher Description= full time Skill= math, science Location= bharatpur Timing= aaaaaaaa Salary= aaaaaaa Category=teacher	Record not added	Record added	Fail

user login testing

Table 4-4 user login testing

ID	Test case description	Test data	Expected result	Actual result	Pass/fail
TC1	user enters an valid email and	Email=ram@gmail.com Pass= 123	user login sucessfull y	As expected	pass

	pass				
TC2	user enters a invalid email and password	Email=ram1@gmail.com Pass:1234	Invalid username or password	As expected	pass

User Registration testing

Table 4-5 user registration testing

ID	Test case description	Test data	Expected result	Actual result	Pass/fail
TC2	user enters an valid email	Name=Aadesh Email=aadesh@gmail.com Pass= 123	Register successfully	As expected	pass
TC2	Admin enters a invalid email and password	Email:234gmail.com Pass:1234	Invalid email address	As expected	Pass

Chapter 5 Conclusion and Future Recommendations

5.1 Lesson Learnt / Outcome

When the project is completed, the user will be able to search their suitable job by looking at the job details. After registration and login to the website they can apply for the job of their choice with required documents. This system is totally free, fast, and reliable website. In this way, users can save money and time from this website.

5.2 Conclusion

After the completion of the online job portal website, people can perform easy apply for the suitable job online. The current website has fulfilled all the objectives. We followed the specifications strictly but enhanced some of the features when there was a need for it to be done. There have been special challenges when it comes to backend and making sure that the website responds in a predictable manner.

The use of PHP in this project is very simple and easy to use. It could handle a lot of data and easily manipulation in compared to another scripting language, this is widely used over the world.

As we came to the end of the project, we realized that there are many enhancements that can be made on the website. Some of the ideas came from those who tested the website and some of them from both of us.

5.3 Future Recommendations

Here is what we can add in the future on this website to increase its usability, user experience and portability of the website. There is a lot to be done hence this website can be considered as a starting point for something big to come. It will need more time and Addition of payment method for register,

Addition of change password categories,

Addition of employer category,

Addition of online interview of selected candidate,

References

Chapter 1

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